Learning Outcome1: Perform Requirements Analysis

Requirements Analysis is the process of identifying, defining, and documenting the needs or requirements of a system, product. It's a critical step in the software development lifecycle (SDLC) that ensures the final product meets the intended goals and expectations.

key steps involved in requirements analysis:

Identify Stakeholders: Determine who will be affected by the project or who has an interest in its outcome. This includes end-users, customers, and other relevant parties

Gather Requirements: Use various techniques such as interviews, surveys, workshops, and observation to collect information from stakeholders about their needs, expectations, and constraints.

1. **Define Requirements**: Clearly articulate the gathered information into specific, detailed, and measurable requirements. This often involves creating documentation such as requirement specifications or user stories.
2. **Analyze Requirements**: Evaluate the requirements to ensure they are feasible, clear, and aligned with the project goals. This may involve assessing their impact, priority, and any potential conflicts.
3. **Validate Requirements:** Ensure that the requirements accurately reflect the stakeholders' needs and that they are achievable within the project's constraints (such as budget, time, and technology).
4. **Manage Requirements**: Establish a process for managing changes to requirements throughout the project lifecycle. This includes tracking changes, updating documentation, and communicating updates to stakeholders.
5. **Document Requirements:** Create comprehensive and organized documentation that captures all requirements and can be used as a reference throughout the project.

- **Introduction to Quality Assurance**
  - ✓ **Describing Quality Assurance (QA)**

Definition: is a systematic process of ensuring that a product or service meets specified quality standards. It involves a series of activities aimed at preventing defects and ensuring that the final product or service is of high quality.

- **Stages/Processes**

1. **Quality Planning:i**s Establish clear quality goals and standards based on customer requirements and project specifications.
2. **Requirements Analysis:** Is a Collect and analyze requirements from stakeholders to ensure they are clear, complete, and feasible.
3. **Design and Development:** Is Integrate quality assurance activities into the design and development phases. This includes reviewing design documents and ensuring adherence to quality standards.

4. **Testing**: Perform various types of testing (e.g., unit testing, integration testing, system testing, acceptance testing) to identify defects or issues.
5. **Defect Management:** Is a Review and analyze defects to determine their root causes.
6. **Quality Control:** Is a Continuously monitor processes and outputs to ensure they meet quality standards.
7. **Review and Feedback**: Collect feedback from stakeholders and end-users to identify areas for improvement.
8. **Documentation and Reporting:** Is a Document QA activity, test results, defect management, and other relevant information. It is Share Findings Communicate findings and recommendations to stakeholders and team members.

- **Main Types of Quality Assurance**

1. **Internal Quality Assurance**: This type of QA is conducted within the organization itself to ensure that products or services meet internal standards and requirements.
   Examples: Internal audits, process reviews, and employee training programs.
2. **External Quality Assurance**: This type of QA involves external organizations or individuals who assess the quality of a product or service.
   Examples: Third-party audits, customer surveys, and independent testing.
3. **Process Quality Assurance:**This type of QA focuses on ensuring that processes are being followed correctly and efficiently.
   Examples: Process audits, process mapping, and process improvement initiatives.
4. **Product Quality Assurance:** This type of QA focuses on ensuring that products meet specified quality standards and requirements.
   Examples: Product testing, inspections, and quality control checks.
5. **Service Quality Assurance:** This type of QA focuses on ensuring that services are delivered effectively and meet customer expectations.
   Examples: Customer satisfaction surveys and service level agreements (SLAs).
6. **Supplier Quality Assurance:** This type of QA ensures that suppliers are providing high-quality products or services.
   Examples: Supplier audits, supplier certification programs, and supplier performance evaluations.

- **Standards:** provide guidelines and benchmark to ensure consistent quality across products, services, and processes.

Key QA standards include:

1. **ISO 14001:** This standard focuses on environmental management, providing guidelines for organizations to minimize their environmental impact.
2. **ISO 27001:** This standard focuses on information security management, providing a framework for organizations to protect their sensitive information.

3. **ASQ Standards:** The American Society for Quality (ASQ) offers a variety of quality standards, including standards for auditing, process improvement, and quality engineering.
4. **IEEE Standards:** Guidelines specific to software engineering, such as IEEE 829 for software test documentation and IEEE 12207 for software lifecycle processes.

- **Methods QA**

- **Inspection:** Reviewing products or processes at various stages to ensure they meet quality criteria. This can involve visual inspections, measurements, and functional tests.
- **Testing:** Conducting various tests (e.g., unit testing, integration testing, system testing) to identify defects and verify that the product performs as expected.
- **Audits:** Systematic reviews of processes and procedures to ensure compliance with quality standards and identify areas for improvement. This includes internal and external audits.
- **Statistical Process Control (SPC):** Using statistical methods to monitor and control processes, ensuring they remain within predefined limits and produce consistent quality.
- **Quality Planning:** Developing detailed plans outlining quality objectives, standards, and processes to guide production and service delivery.
- **Customer Feedback:**is a Collecting feedback from customers to assess satisfaction levels.It is Conducting focus groups to gather in-depth feedback from customers.

- **Benefits of Quality Assurance**

- ✓ **Improved customer satisfaction:** Products and services that meet or exceed customer expectations lead to higher satisfaction levels.
- ✓ **Reduced costs:** Preventing defects and rework can significantly reduce costs associated with production, rework, and customer complaints.
- ✓ **Increased efficiency:** Effective QA processes can improve efficiency and productivity by identifying and eliminating waste.
- ✓ **Competitive advantage:** High-quality products or services can provide a competitive advantage in the marketplace.
- ✓ **Risk management:** QA can help identify and mitigate risks associated with product or service quality.
- ✓ **Employee morale:** A focus on quality can improve employee morale and motivation.
- ✓ **Continuous improvement:** QA can drive continuous improvement efforts, leading to ongoing enhancements in products, services, and processes.

- ✓ **Classification of quality assurance**
  - **Industry-Specific QA**: refers to quality assurance practices tailored to the unique requirements and challenges of a particular industry. Different industries have specific standards, regulations, and customer expectations that must be met to ensure product or service quality examples: Healthcare, Manufacturing and Financial services
  - **Process and Product QA**

Process QA: focuses on improving and verifying the methods and procedures used in production or service delivery to ensure they consistently produce high-quality results. It involves assessing and refining workflows, training, and compliance with established standards to enhance overall efficiency and effectiveness.

Product QA: on the other hand, is concerned with the final output—whether it's a product or service. It involves testing and evaluating the product against quality standards and specifications to ensure it meets customer expectations and is free of defects before it reaches the market.

In essence, Process QA aims to optimize the creation process, while Product QA ensures the end result meets quality requirements.

- **Internal vs. External QA**

Internal QA refers to quality assurance activities carried out within an organization by its own employees or teams.

while

External QA involves quality assurance activities carried out by independent, third-party entities consultants who are not part of the organization

- **Manual vs. Automated QA**

Manual testing can be used to identify exploratory defects and provide insights into user experience, **while automated testing** can be used to efficiently execute repetitive tests and improve test coverage.

Feature	Manual QA	Automated QA
Execution	Human testers manually execute test cases.	Automated tools execute test cases based on scripts.
Speed	Slower	Faster
Efficiency	Can be error-prone and time-consuming	Can be more efficient and reduce manual effort.
Coverage	Can identify subtle defects	May miss certain types of defects.
Cost	Lower upfront cost, but can be expensive in the long run.	Higher upfront cost for tools and expertise, but can be cost-effective in the long run.
Maintenance	Requires ongoing maintenance and updates.	Requires ongoing maintenance and updates, but automation can simplify the process.
Suitability	Suitable for exploratory testing, usability testing, and ad-hoc testing.	Suitable for regression testing, performance testing, and repetitive tasks.

- ✓ **Quality Assurance Vs Quality Control**

QA is a proactive process that focuses on preventing defects and ensuring that quality is built into the product or service from the beginning.
QC is a reactive process that focuses on identifying and correcting defects after they occur.

Feature	Quality Assurance	Quality Control
Focus	Prevention of defects	Detection and correction of defects
Timing	Proactive	Reactive
Scope	Organization-wide	Specific products or processes
Activities	Planning, standardization, process improvement	Inspection, testing, corrective actions

In summary, QA is about building quality into the process, while QC is about checking quality in the final product. Both are essential for delivering high-quality products and services.

- ✓ **Identify tools and techniques**

Identify tools of Quality Assurance

- Flowchart. A flowchart could be a diagram representing a workflow method, or a step-by-step method to connect by arrows and lines in several directions.
- Histogram.
- Check Sheet.
- Pareto Chart.
- Control Chart.
- Scatter Diagram

- ✓ **Techniques of Quality Assurance**

Quality assurance (QA) involves various techniques to ensure that products or services meet specified quality standards.

- **Inspection and Testing:**Is Verifying or Evaluating the speed, efficiency, and reliability of products or services and Ensuring that products or services meet customer requirements before delivery.
- **Statistical Process Control (SPC):** is a Selecting representative sample for inspection or testing. It is Evaluating whether a process is in control or out of control.
- **Quality Management Systems (QMS): Is a** Maintaining comprehensive documentation of quality activities and processes, conducting regular audits to assess compliance with quality standards.
- **Total Quality Management (TQM):**is a Putting the customer at the center of all activities, encouraging employee participation in quality initiatives
- **Customer Feedback**:is Collecting feedback from customers to assess satisfaction levels. And Addressing customer complaints promptly and effectively.